

An Insulator in Transmission Lines Recognition and Fault Detection Model Based on Improved Faster RCNN

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Abstract—Insulators are critical electric components in transmission lines. Recognizing insulators and detecting the faults timely and accurately is essential for maintaining the safety and stability of transmission lines. Traditional methods have low accuracy and poor applicability in insulator recognition and fault detection. An insulator recognition and fault detection model was proposed in the article aiming at improving the insulator recognition and fault detection accuracy. First, based on the faster region convolutional neural network (RCNN), the feature pyramid networks (FPNs) were used to improve the Faster RCNN model and locate the insulators with complex background image. Then, the target area was clipped to remove the redundant background noise, and the hue, saturation, and value (HSV) color space adaptive threshold algorithm was applied for image segmentation due to the influence of light, background noise, and shooting angle. Finally, line detection, image rotation, and vertical projection were used to finish the insulator fault detection. The experimental results show that the proposed insulator recognition and fault detection model can recognize the insulators and detect fault types with better accuracy and achieve a mean average precision (mAP) of 90.8% for glass insulators and 91.7% for composite insulators on the testing dataset. Additionally, the proposed method meets the intelligent inspection of insulator faults in transmission lines and has good engineering application value.

Index Terms—Faster region convolutional neural network (RCNN), fault detection, feature pyramid network (FPN), hue, saturation, and value (HSV), insulator.

I. INTRODUCTION

INSULATORS, as an essential part of the transmission lines, work for a long time in harsh environments such as strong electric field, ultralow temperature, freezing rain, and snow.

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The fault insulators will threaten the safe operation of the transmission lines or even cause substantial economic losses in some extreme cases. Because of this, it is vital to identify the fault insulators and determine the fault type for the safe operation of the whole system.

The premise of insulator fault detection is the accurate recognition of insulator targets. Traditional insulator recognition methods include threshold segmentation based on target color features [1] and location based on contour features [2]. However, these methods need to design the target features manually, and different target scenarios require different features, which are of low recognition accuracy and may lead to missed or false detection caused by operators' subjective factors. There are some other methods to detect insulator faults. Ha *et al.* [3] proposed a hierarchical fault detection method for transmission lines using a microphone array to locate the faults and thermal imaging and charge coupled device (CCD) cameras to verify the fault and store the image, respectively. But this method cannot accurately detect the insulator faults, and the use of sound patterns to locate the insulator is easily disturbed by external noise, resulting in poor detection results. A saliency and adaptive morphological-based insulator faults detection algorithm was proposed in [4], and it achieved 92% accuracy on average. However, the method can only detect the bunch-drop fault of the glass insulator. Zhai *et al.* [5] proposed an algorithmic solution for the bunch-drop fault detection for both glass and ceramic insulators based on spatial morphological features. The average fault detection success rate in ceramic insulators can reach over 91%, while glass insulators' average fault detection success rate can reach over 92%. However, the automatic differentiation of insulator types is not implemented, and the size of the fault area in insulators cannot be accurately obtained during the fault detection process.

Compared with traditional methods, the convolutional neural network (CNN) can automatically extract more remarkable features from images, reduce the recognition time, and improve the accuracy. In [6], CNN was used to realize the automatic positioning of insulators, and the accuracy of insulator location had reached 90.5%. However, it cannot complete the fault detection of insulators. Zhao *et al.* [7] introduced the adversarial spatial dropout network layer, and the average accuracy of insulator recognition was improved to reach 84.29%, but the accuracy is lower, and there is still room for improvement. Liu *et al.* [8] proposed an

insulator recognition and fault detection method composed of a convolution network, a region proposal network (RPN), and a target detector. The technique only detected one type of insulator faults, and the precision for missing caps fault reached 92%. Wang *et al.* [9] proposed an automatic insulator diagnosis method using instance segmentation and temperature analysis of infrared insulator images. The mean average precision (mAP) reached 77%. However, there is still a lot of room for improvement, and the method is applied to infrared insulator images. It cannot be applied to the red, green, and blue (RGB) three-channel color images.

The methods above were developed based on handcrafted features or automatic learning techniques and can recognize insulators and detect faults of insulators under specific detection conditions. But in most insulator images, the background image usually includes various scenarios such as forests, rivers, farms, and land. The number of pixels containing insulator information in the whole image is small. Therefore, it is particularly important to choose a target detection algorithm that can fully extract the insulator target information from images. Faster region CNN (RCNN) has higher recognition accuracy and faster computing speed than other deep learning object recognition methods. A large number of references were written based on the Faster RCNN algorithm for insulator recognition and fault detection. In [10], the Faster RCNN (Region CNN) framework was used to recognize the insulators and achieved good recognition results that the accuracy reached 90.5%. However, the effect is not ideal for situation where there is obscuration between the insulators. Tao *et al.* [11] adopted a cascaded insulator fault detection model. The defect detection precision and recall of the proposed method in [11] were 91% and 96%, respectively, using the standard insulator dataset. But its insulator recognition and fault detection part are two separate networks, and the model is more complicated, so the cost is more. In [12], the proposed method adopted the self-shattering location by combining Faster RCNN with U-net. The value of precision was 95.1%, and the value of recall was 95.5%. But it did not combine detection and segmentation in the model, so it took more time to train the two parts separately. Lei and Sui [13] proposed a deep convolution neural network method based on the Faster RCNN method to locate the broken insulators and bird nests. The value of average precision of the three classes is 97.6%, and the average recall is 97.9%. However, the network training image data are small, only 448 images, which may lead to insufficient training of the network model. Also, the method uses Resnet101 as the feature extraction network, resulting in a much slower detection speed. Zhao *et al.* [14] put forward an insulator inspection image based on the improved Faster RCNN to detect insulators. The method proposed in [14] reached 81.8% accuracy for insulator detection. But the method cannot distinguish different types of insulators. Zheng *et al.* [15] proposed a smart detection method of power transmission line based on RCNN. The accuracy of standard insulators was 91%, and the accuracy of fault insulators was only 74%. The detection accuracy of fault insulators is lower. The studies discussed above have carried out work on insulator recognition and fault detection based on

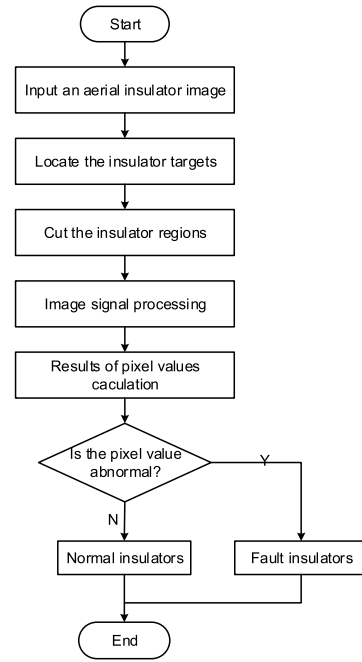


Fig. 1. Flowchart of the proposed fault detection method.

the Faster RCNN method from different angles and achieved specific positive results. However, fault detection is of a single type, and some methods can only complete the fault location. There is still room for improvement in the detection accuracy of insulator recognition and fault detection.

The work in the article focuses on the detection of multiple types of faults based on the completion of insulator recognition, aiming at improving the accuracy of insulator recognition and fault detection. Given this, we propose an insulator recognition and fault detection model based on the improved Faster RCNN. The article improves the Faster RCNN network and uses it as the insulator located network, accurately recognizing insulators in the images. Considering the shape of the insulator target, image signal processing is used to detect the faults of the recognized insulators, which can precisely detect crack and bunch-drop faults of the glass and composite insulators, and the time cost is relatively more minor. The rest of this article is organized as follows: In Section II, the recognition model of Faster RCNN based on feature pyramid networks (FPNs) is introduced to identify the insulator automatically and effectively. In Section III, the model of insulator fault detection is introduced, and the adaptive threshold segmentation and line detection are applied to detect the insulators with different faults. Section IV gives the experiment and result from analysis.

II. INSULATOR RECOGNITION MODEL

The flowchart of insulator fault classification is depicted in Fig. 1. First, insulator image is put into the improved Faster RCNN network to locate the insulator targets and clip the insulator regions. Next, image signal processing is performed, which includes adaptive threshold segmentation, line detection, image level correction, and pixel value calculation.

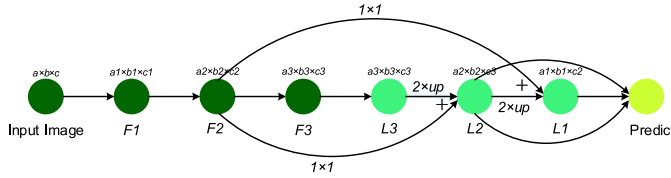


Fig. 2. FPN end-to-end signal flow diagram.

Finally, according to the results of pixel value calculation, the fault classification of the insulator targets is carried out, and the classification result is output.

To enhance the insulator recognition and fault detection accuracy, we make some improvements in Faster RCNN, which is composed of RPN network and Fast RCNN. The convolution network extracts the image features and generates the feature map. After that, the feature map is input into RPN to generate the feature information of the candidate box. Finally, the Fast RCNN algorithm is used to classify the feature information generated by RPN. Due to space limitation, the readers can refer to [16] for details. When Faster RCNN is used for target detection, the insulator targets are recognized in the last layer. For smaller scale insulators, the semantic information loss becomes severe, which might cause missing detection of smaller scale insulators and affect the insulator recognition accuracy. To solve this problem, the article introduces the FPN.

A. FPN Model

The FPN model uses the bottom-up pathways, top-down pathways, and lateral connections to combine the high-level features of low-resolution and high-semantic information with the low-level features of high-resolution and low-semantic information so that the features at each scale have rich semantic information [17]. The end-to-end signal flow diagram of the FPN is depicted in Fig. 2.

The bottom-up path is a part of the feedforward feature extraction network. After multiple convolution operations, the size of feature maps is continuously reduced, and the abstract semantic information contained therein is also continuously increased. For the top-down path, the high-level feature maps which contain rich abstract semantics information are first up-sampled to the same scale as the feature maps of the previous layer. Through the horizontal connection, it is added to the feature maps of the previous layer to perform feature fusion to realize the downward transmission of the abstract semantic information of high-level feature maps. As a result, the semantic information of each layer of feature maps below has been enhanced.

As shown in Fig. 2, the bilinear interpolation is used to double up-sample the feature maps L3 and L2 to make their size the same as the feature maps of the previous layer. The indicator “ $2 \times \text{up}$ ” means that the resolution of the expanded feature map is twice that of the original one through up-sampling. Then the feature maps extracted from the bottom-up of the backbone network are F1, F2, and F3. Suppose their scales are $a_1 \times b_1 \times c_1$, $a_2 \times b_2 \times c_2$, and $a_3 \times b_3 \times c_3$, and the scale of the input images is $a \times b \times c$. L3 comes

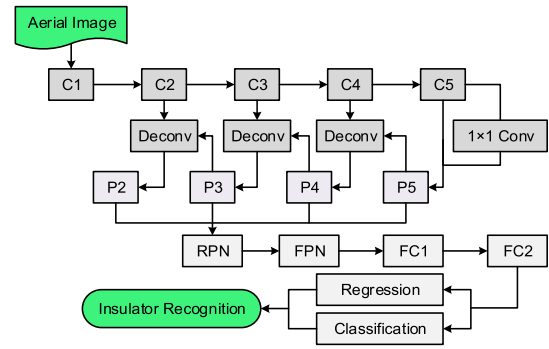


Fig. 3. Flowchart of the proposed insulator recognition method.

from feature map F3, and the scale is $a_3 \times b_3 \times c_3$. For the feature map L2, first up-sample L3 is doubled to change the scale to $a_2 \times b_2 \times c_3$. Then the number of channels of F2 to c_3 through 1×1 convolution is adjusted, and the scale of F2 to $a_2 \times b_2 \times c_3$ is converted. Finally, the two are added and ablated by 3×3 convolution to obtain the feature map L2. The generation process of the feature map L1 is similar to L2. The process is repeated until the most refined feature map is produced. The mathematical expression is as follows: “ $\text{up}_{\times 2}$ ” represents the double up-sampling, and “ $C_{1 \times 1}$ ” represents the 1×1 convolution operation:

$$\begin{aligned} L3 &= F3 \\ L2 &= \text{up}_{\times 2}(L3) + C_{1 \times 1}(F2) \\ L1 &= \text{up}_{\times 2}(L2) + C_{1 \times 1}(F1). \end{aligned} \quad (1)$$

To obtain better feature map fusion information, the up-sampling method of interpolation and 1×1 convolution is replaced with deconvolution. The deconvolution is used to up-sample the feature maps of the previous layer to keep the dimension consistent. Compared with the bilinear interpolation up-sampling method adopted by the original FPN, deconvolution is acquired through network training, and better details can be obtained through fusion after deconvolution to obtain as reliable semantic information as possible.

In this article, the proposed algorithm introduces FPN after extracting the feature information of proposals in RPN to enrich the semantic information. The proposals with significant differences scale are mapped to the feature map region. Each feature map region is then equally divided and the max-pooling operation is performed. The max-pooling feature map is input into the FPN to generate insulator recognition feature maps of different scales, and then through the whole connection layer, the classification and regression are completed, and finally, the insulator is identified.

B. Insulator Recognition by Improved Faster RCNN

As the Faster RCNN only uses the feature map generated by the highest layer of the feature extraction network to locate the targets, the insulator’s images’ recognition accuracy with smaller sizes in the insulator image is not satisfactory [16]. Therefore, the FPN improved Faster RCNN model with deconvolution is put forward in this article to get more semantic information, and Fig. 3 shows the flowchart.

For the FPN, first, a pyramid hierarchy is defined for each stage. The last layer output of each stage is used as a reference set for the feature map. For the VGG16 [18] feature extraction network, the feature reference set is represented as {C2, C3, C4, C5}, corresponding to the output feature maps of the Conv2, Conv3, Conv4, and Conv5 layers in the VGG16 network, and the side lengths are, respectively, 1/4, 1/8, 1/16, and 1/32 of the side length of the input image. Because of the large memory footprint, the pyramid network did not include Conv1. The 1×1 convolutional kernel layer is appended after C5 to produce the roughest resolution map. Finally, the 3×3 convolution is appended to each merged map, producing the final feature map to reduce the aliasing effect of the up-sampling. The final feature map set is {P2, P3, P4, P5}, corresponding to {C2, C3, C4, C5} with the exact spatial sizes.

The insulator recognition process based on the improved Faster RCNN can be divided into the following steps.

Step #1: Image preprocessing. To input the original image, first, the image size should be limited. The Faster RCNN network limits that the shorter edge of the input image is no more than 600, while the longer one should be no more than 1000, and priority is given to the limitation of the longer edge.

Step #2: Feature pyramid generation. A CNN is used to extract the target features in the image, and the RPN and the detection network share the generated feature map. The VGG16 feature extraction network is adopted; meanwhile, the FPN with deconvolution replaces the single-scale feature map with RPN. A single-scale anchor point is assigned at each level of the feature pyramid, and the corresponding anchor points of {P2, P3, P4, P5} are $\{64^2, 128^2, 256^2, 512^2\}$, while the ratio of three kinds of length to width is $\{1:1, 1:2, 2:1\}$. Therefore, there are 12 anchor points in the pyramid.

Step #3: Loss function calculation. Input the four feature maps obtained in step 2 into RPN for layer-by-layer processing. After convolution, pooling, and other operations, a certain mapping relationship exists between the generated feature map and the input image in terms of size. Therefore, the position of each point on the feature map can be found corresponding to the input image, and the sliding window strategy is applied to generate a large number of anchors in this position. We classify each anchor as the background or foreground and then correct the foreground anchor using the bounding box regression to get the proposals. The nonmaximum suppression [19], [20] is used to sort these proposals according to the confidence scores, then the redundant proposals are removed, and the proposals with higher scores are selected as the output of regions of interest (RoIs). During the training, the loss function of RPN is obtained by the following equation:

$$L(\{p_i\}, \{t_i\}) = \frac{1}{N_{\text{cls}}} \sum_i L_{\text{cls}}(p_i, p_i^*) + \lambda \frac{1}{N_{\text{reg}}} \sum_i p_i^* L_{\text{reg}}(t_i, t_i^*) \quad (2)$$

where i is the index corresponding to the anchor in the small-batch data; p_i is the probability that the anchor is predicted

as the target. If the anchor is a positive sample, p_i is 1; otherwise, p_i is 0. λ represents the balance parameters. L_{cls} represents the binary classification loss and L_{reg} represents the loss of position regression which are normalized by N_{cls} and N_{reg} ; N_{cls} is the mini-batch size, and N_{reg} equals to the number of anchors. t_i is the parameterized coordinate vector of the predicted anchor and t_i^* is the coordinate vector of ground truth corresponding to the anchor in the foreground.

Step #4: FPN feature layer selection and RoI pooling. The shape of the target box extracted in Step 3 is quite different, making the corresponding regions on the feature map vary. Therefore, the feature map of each region is divided into the same size blocks, and max-pooling is performed on each block. The size of the feature map of all the regions is fixed as 7×7 . The Fast RCNN uses the RoI pooling layer to extract features, which is usually performed on a single-scale feature map. To combine the Fast RCNN with the FPN, RoI should be assigned at different scales to the pyramid scale. For the RoI with width w and height h , the level k of the corresponding feature pyramid is determined by the following equation:

$$k = \lfloor k_0 + \log_2(\sqrt{wh}/224) \rfloor. \quad (3)$$

The pretrained image size of the ImageNet dataset is 224, and k_0 is the target level of RoI mapped to $w \times h = 224 \times 224$. We set k_0 to 4.

Step #5: Results obtained. Input the feature map in Step 4 and calculate the probability of identifying each proposal box as a target, and the bounding box regression is used to fine-tune the target box, and the recognition results are obtained [21].

III. FAULT INSULATOR DETECTION

A. Adaptive Threshold Segmentation

The RGB color space does not follow the subjective judgment of people of color similarity, because it does not use the distance between two colors to represent the perceived differences between two kinds of color. Hue, saturation, and value (HSV) color space is more by people's subjective cognition, so we add RGB images into the HSV image first; the process of converting RGB into HSV is as follows.

Step #1: Divide the RGB value corresponding to the picture by 255 to get R' , G' , and B' by the following equation:

$$R', G', B' = R, G, B/255. \quad (4)$$

R , G , and B represent 0–255 levels, a total of 256 levels of brightness.

Step #2: Calculate the difference between the maximum value of R' , G' , and B' and the minimum value and record it as Δ

$$\Delta = \max(R', G', B') - \min(R', G', B'). \quad (5)$$

Step #3: Calculate the corresponding H, S, and V color gamut values by (6)–(8) to obtain the corresponding

HSV image

$$H = \begin{cases} 0^\circ, & \Delta = 0 \\ 60^\circ \times \left(\frac{G' - B'}{\Delta} + 0 \right), & \max(R', G', B') = R' \\ 60^\circ \times \left(\frac{B' - R'}{\Delta} + 2 \right), & \max(R', G', B') = G' \\ 60^\circ \times \left(\frac{R' - G'}{\Delta} + 4 \right), & \max(R', G', B') = B' \end{cases} \quad (6)$$

$$S = \begin{cases} 0, & \max(R', G', B') = 0 \\ \frac{\Delta}{\max(R', G', B')}, & \max(R', G', B') \neq 0 \end{cases} \quad (7)$$

$$V = \max(R', G', B'). \quad (8)$$

Then for the glass insulators and composite insulators, the corresponding hue, saturation, and brightness are fit, respectively. According to the numerical interval and removing the effects of light and noise, we use median filtering and adaptive threshold segmentation. The pixels satisfying the value range are reassigned to 255, while the other pixels are reassigned to 0; thus, the binary image after segmentation can be obtained.

B. Line Detection

Due to the different angles of the insulator images, if the pixel values of each column of the binary image are counted directly, the pixel values of each column with inclination angle cannot correspond to the actual coordinates one by one [22]. According to the linear characteristics of insulators in the image, find out all possible straight lines ($L1, L2, \dots, Ln$) in the image by Hough transform line detection algorithm, and take the average of the inclination angles of all straight lines to obtain the inclination angle of the insulator target.

The steps of the Hough transform line detection are as follows.

Step #1: Canny edge detection. There are six steps in all, including original image graying, image Gaussian filtering, gradient amplitude, direction calculation, nonmaximum suppression of gradient amplitude, detection, and edge connection with a double threshold algorithm [23].

Step #2: Conversion of the pixel points detected by the edge from the plane space to the polar coordinate space. If different pixel points are on a straight line, there will be an intersection point in the polar coordinate space. Therefore, to get a straight line in the image, the distance from the intersection point to the pole and the angle between r and the x -axis can be obtained by (9). According to (10), the inclination angle of the line is D

$$y = \left(-\frac{\cos \theta}{\sin \theta} \right) x + \left(\frac{r}{\sin \theta} \right) \quad (9)$$

$$D = \arctan \left[\sum_i^n (K_i) / n \right] \quad (10)$$

where k_i is the slope of ($L1, L2, \dots, Ln$).

C. Image Level Correction

According to the inclination angle D of the insulator obtained by (4), the image is rotated to keep the insulator

horizontal to facilitate calculating the insulator pixel values on the binary image.

D. Pixel Values Calculation

Sum the pixel values of each column of the rotated image. The pixel value calculation results are plotted in Fig. 6(g). The horizontal axis represents each column of the rotated image, and the vertical axis represents the summed pixel value.

E. Detection of Fault Insulator

In view of the bunch-drop fault of glass insulators, the statistical curve chart of the pixel value of the insulator image is analyzed, recounted as follows.

- 1) For a standard-type insulator, the curve fluctuates regularly without abnormal fluctuations.
- 2) For an insulator with a bunch-drop fault, the abnormal fluctuation of the pixel value appears on the graph below half the peak value, and the x -coordinate range of the fluctuation position is more significant than threshold T1.
- 3) For an insulator with a crack fault, the image is first detected by the Canny operator for edge detection, then the crack region is extracted by the adaptive threshold segmentation algorithm, and finally the crack characteristics are extracted for distinction [24]. Two characteristic parameters, the length and the crack area, are selected to detect the insulator crack fault. According to the projected length of the crack region in the horizontal direction x and the projected length in the vertical direction y , the crack length l can be obtained by the following equation:

$$l = \sqrt{x^2 + y^2}. \quad (11)$$

The crack can be defined as transverse crack by the following equation:

$$0^\circ \leq \arctan\left(\frac{y}{x}\right) \leq 45^\circ. \quad (12)$$

The crack can be defined as vertical crack by the following equation:

$$45^\circ \leq \arctan\left(\frac{y}{x}\right) \leq 90^\circ. \quad (13)$$

By counting the number of black pixels on the edge, we can get the characteristics of the crack region. For a characteristic parameter of the length larger than threshold T2 and a characteristic region parameter larger than threshold T3, the fault type is determined to crack. For a characteristic parameter of the length less than threshold T2 or the characteristic region parameter larger than threshold T3, the insulator state is normal.

IV. EXPERIMENT AND RESULT ANALYSIS

The operating system used is Ubuntu 16.04 LTS, and the graphic processing unit (GPU) is NVIDIA GeForce GTX 1080Ti; the framework for deep learning is TensorFlow 1.3.0.

TABLE I
INSULATOR DATASET

	Index	Dataset	Training	Testing	Total
Glass	#1	insulator	2425	809	3234
Insulator	#2	normal + crack + bunch-drop	1734+215 +476	566+97 +146	3234
Composite	#3	insulator	1500	500	2000
Insulator	#4	normal + crack +bunch-drop	1181+114 +205	394+38 +68	2000

A. Dataset and Experiment

The source of the dataset has two parts; one part is provided by Tao *et al.* [11], which contains 248 defective composite insulator pictures and 600 normal composite insulator pictures. The other part is collected and selected by ourselves, which is taken on the transmission lines through the camera mounted on the drone. The size of the images in the dataset is 1152×864 uniformly.

The training set and the test set are randomly divided according to the ratio of 7:3. The dataset division is presented in Table I. The threshold values of experimental parameters T1, T2, and T3 were set as 20, 200, and 350, respectively. Datasets #1 and #2 are the glass insulator datasets, which are completely collected and selected by ourselves. There are 3234 glass insulator images in both datasets. Dataset #2 includes 622 bunch-drop images and 312 crack images. Datasets #3 and #4 are the composite insulator datasets, which include 2000 composite insulator images in both datasets. There are 848 composite insulator pictures from the literature [11], and the remaining 1152 pictures are collected and selected by ourselves. Dataset #4 includes 273 bunch-drop images and 152 crack images.

The experiment in this article is mainly divided into two parts. The first part uses datasets #1 and #3 in Table I for experiments to verify the effectiveness of the insulator recognition network proposed in the article. There is only one class in the datasets, named insulator. The second part uses datasets #2 and #4 for experiments to verify the fault detection effect of glass insulators and composite insulators; there are three classes in the datasets, named normal, bunch-drop, and crack, respectively. For a picture containing multiple insulators, as shown in Fig. 6, the picture contains a normal insulator and a bunch-drop insulator. We judge it as a defective picture and classify it as a bunch-drop category.

When training the network, first, set the learning rate to 0.001 and train 30 000 iterations; then, set the learning rate to 0.0001 and train 10 000 iterations. The curve of total loss changes that model trained 40 000 iterations is shown in Fig. 4. When iterated 40 000 times, the model becomes stable.

B. Recognition and Fault Detection for Glass and Composite Insulators

The test results of glass insulators of bunch-drop are given in Figs. 5 and 6. Through the improved Faster RCNN network model, the insulator targets in the original glass insulator image were recognized, and all the targets were labeled with boxes. After that, the insulator image was clipped along

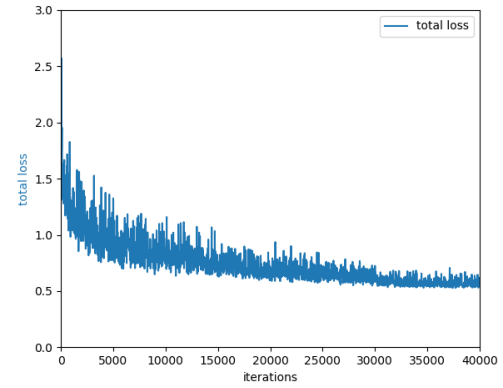


Fig. 4. Total training loss change curve.

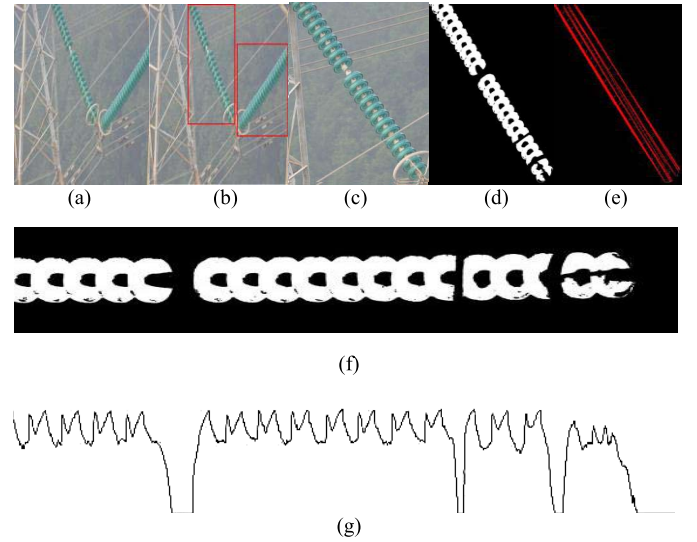


Fig. 5. Process of bunch-drop fault detection. (a) Original glass insulator image. (b) Located insulator image. (c) Cutting boundary image. (d) Segmentation image. (e) Line detection image. (f) Image level correction image. (g) Calculate pixel image.

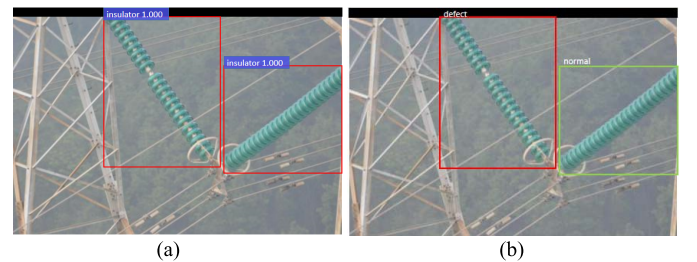


Fig. 6. Results of bunch-drop fault detection. (a) Original glass insulator recognition image. (b) Original glass insulator bunch-drop fault detection image.

the boxes, and thus a part of the image containing the insulator only was obtained. The obtained image was clipped by the adaptive threshold based on the HSV color space and then line detection was made. Finally, the latest image was horizontally corrected, and the pixel value was counted.

If the abscissa range in the abnormal region is more significant than threshold T1, the fault type is judged to be bunch-drop; otherwise, the crack should be analyzed further.

1) *Recognition for Insulators:* For method validation and comparison, the improved algorithm proposed in this article, you only look once (YOLO)v2 [25], single-shot multibox

TABLE II
RESULTS OF CLASSIFICATION NETWORK ON DATASETS #1 AND #3

Algorithms	mAP1(%)	mAP2(%)
<i>YOLOv2</i>	88.9	88.3
<i>SSD</i>	89.3	88.6
<i>Faster RCNN</i>	90.1	90.3
<i>Literature [4]</i>	91.8	91.2
<i>Literature [5]</i>	92.3	91.6
<i>Proposed Algorithm</i>	92.6	93.0

detector (SSD) [26], Faster RCNN, and the methods proposed in the literature [4] and [5] were trained and tested on datasets #1 and #3. The mAP was used as the evaluation index. The results are listed in Table II. The mAP1 represents glass insulators, and mAP2 represents composite insulators.

According to the detection results of the insulator locating network in Table II, the proposed insulator recognition algorithm has the highest accuracy for glass insulators and composite insulators compared with the algorithms listed in Table II. This is because the two-stage target detection algorithm Faster RCNN has better target detection effect. Moreover, the method in this article adds an improved FPN structure to the Faster RCNN network, which increases the feature fusion between feature maps. It can make the network better to extract the insulator target feature. Therefore, the insulator recognition effect is better.

2) *Fault Detection of Glass Insulators*: For validation, the proposed fault detection algorithms are applied on dataset #2 and compared with method 1, method 2, and the methods presented in the literature [11], [13], and [15]. Method 1 is the improved Faster RCNN insulator detection based on VGG16, and method 2 is the improved Faster RCNN insulator detection based on Resnet34. The results are listed in Table III. The article also used the mean absolute error (MAE) as the evaluation index.

In Table III, the results show that the proposed algorithm has the highest fault detection accuracy for glass insulators. Compared with method 1, method 2, and the methods in [11], [13], and [15], the fault detection accuracy is higher, respectively, 36.8%, 28.4%, 0.7%, 21.8%, and 5.2%. It can be observed that the proposed method outperforms better with significantly improved fault detection accuracy and the lowest MAE value.

The reason lies that after the proposed algorithm accurately identifies the insulator targets, the designed image signal processing method is used to further detect the insulator fault. It can eliminate a large amount of noise through insulator recognition and segmentation steps and focuses more on the subtly different morphological characteristics among normal insulators, crack insulators, and bunch-drop insulators. So, the fault detection effect is better.

However, shielding such as tower poles and wires greatly influences the judgment of normal insulators. Therefore, in this case, a normal insulator may be recognized as a fault insulator.

3) *Fault Detection of Composite Insulators*: The fault detection algorithm was applied on datasets #4 and compared with method 1, method 2, and the methods in [11], [13], and [15], respectively. The results are listed in Table IV.

TABLE III
RESULTS OF FAULT DETECTION NETWORK ON DATASET #2

Algorithms	Normal (%)	Bunch-drop (%)	Crack (%)	mAP (%)	MAE (%)
<i>Method 1</i>	49.6	61.4	50.9	54.0	46.0
<i>Method 2</i>	60.1	67.7	59.3	62.4	37.6
<i>Literature [11]</i>	88.5	90.3	91.6	90.1	9.9
<i>Literature [13]</i>	70.3	75.1	61.7	69.0	31.0
<i>Literature [15]</i>	87.6	85.6	83.7	85.6	14.4
<i>Proposed Algorithm</i>	88.7	91.2	92.5	90.8	9.2

TABLE IV
RESULTS OF FAULT DETECTION NETWORK ON DATASET #4

Algorithms	Normal (%)	Bunch-drop (%)	Crack (%)	mAP (%)	MAE (%)
<i>Method 1</i>	50.3	64.0	48.3	54.2	45.8
<i>Method 2</i>	61.8	70.3	59.1	63.7	36.3
<i>Literature [11]</i>	88.3	90.7	92.0	90.3	9.7
<i>Literature [13]</i>	71.1	79.6	62.4	71.0	29.0
<i>Literature [15]</i>	86.7	86.3	85.2	86.1	13.9
<i>Proposed Algorithm</i>	88.9	91.3	94.9	91.7	8.3

TABLE V
MEMORY COST, DETECTION TIME, AND TIME COMPLEXITY OF THE METHODS

Methods	Memory cost (GB)	Detection time(s)	FLOPs
<i>Method 1</i>	8.5	0.098	2.1×10^{10}
<i>Method 2</i>	7.8	0.061	9.1×10^9
<i>Literature [11]</i>	10.6	0.425	3.2×10^{10}
<i>Literature [13]</i>	9.7	0.203	1.5×10^{10}
<i>Literature [15]</i>	6.5	0.054	4.3×10^9
<i>Proposed Algorithm</i>	9.3	0.197	2.4×10^{10}

The results of fault detection for composite insulators are listed in Table IV. It shows that the proposed algorithm has better detection accuracy and lower MAE value than the methods listed in Table IV. Therefore, the superiority of the proposed method is demonstrated for fault detection of both composite insulators and glass insulators.

4) *Memory Cost, Detection Time, and Time Complexity*: To better prove the advantages of the proposed algorithm, we have also compared the memory cost, detection time, and time complexity with other methods. Memory cost is the GPU memory requirement for the methods when running in peak memory requirement. Detection time represents the time it takes to detect one picture. Floating point of operations (FLOPs) are the number of floating-point operations, which can be used to measure the algorithm's complexity.

Moreover, memory cost, detection time, and time complexity of the methods are given in Table V. It can be found that compared with other methods, it consumes less detection time, which means faster detection of insulator faults. After recognizing the insulator targets, the image signal processing method is used in insulator fault detection, which does not need to go through a complex neural network to detect faults. Therefore, the detection speed is faster and the latency is smaller. Although the number of FLOPs is enormous and the cost of GPU memory is relatively high, when the hardware

resources are satisfied, the method proposed in this article can detect insulator faults faster and more accurately. It can satisfy the requirement of engineering applications.

V. CONCLUSION

In this article, we proposed an improved Faster RCNN algorithm for insulator recognition and fault detection, which is a combination of Faster RCNN and improved FPN. In terms of insulator recognition, the proposed algorithm shows significant superiority over traditional algorithms. The fault detection results are more accurate in comparison to other methods. Moreover, our algorithm is applicable for both bunch-drop and crack fault detection. It can be concluded from the results that our algorithm is of great advantage in recognizing and detecting fault insulators in transmission lines. In future work, we will be committed to improving the computing speed of the algorithm and applying it to practical engineering.

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